A Cross-sectional AF-duration analysis in GSE41177

Descriptive chronicity proxy; not recurrence, progression, or treatment response.

REFERENCE

Frozen NCP from Fig 1
30 programs → 16 families
Frozen LE sets
Frozen program→family map

DATA & DURATION

GSE41177 · Affymetrix
38 samples · LAA + LAPV
Month-coded AF duration

Primary:
12 AF-LAA samples
with finite numeric months

Parsing: XM→X · 0M→0
nonnumeric→NA
No year-to-month conversion

FAMILY ACTIVITY

Exact-reference rank scoring
Program-level scores
→ equal-weight family activity

Not aCAIrel
No paired LA–RA contrast

ANALYSIS

PRIMARY LAA · 12 AF-LAA
Spearman ρ · bootstrap 95% CI · permutation p
Class from |ρLAA| only
Strong |ρ| ≥ 0.4
Weak 0.2 ≤ |ρ| < 0.4
Stable |ρ| < 0.2

LAPV REGIONAL SENSITIVITY
regional Spearman ρ · not for class
SR excluded from ρ, p, and class

B Family-level duration trends across atrial regions

Primary LAA associations and LAPV regional sensitivity
16 frozen NCP families · LAA primary: 12 AF samples · Spearman ρ

			LAA Primary n = 12	LAPV Regional sensitivity n = 12	Regional sign
LA	F1	Translation / RQC	0.1261	0.2802	Same
LA	F2	Mito Ca^{2+} transport	-0.5639	0.1961	Opp.
LA	F3	Adipo–lipid axis	0.3257	0.1191	Same
LA	F4	Mito proteostasis / FA–BCAA	-0.0280	-0.1821	Same
LA	F5	tRNA aminoacylation	0.1226	-0.0841	Opp.
LA	F6	Mito translation / import	0.1121	-0.1891	Opp.
LA	F7	OXPHOS / ETC	-0.2977	-0.1751	Same
LA	F8	Complex I biogenesis	0.2452	-0.1261	Opp.
LA	F12	rRNA processing	0.0280	0.5849	Same
LA	F13	Amino-acid metabolism	0.2347	-0.0525	Opp.
LA	F14	Translation initiation	0.1471	0.5919	Same
RA	F9	HS-GAG / bone / GPCR	-0.4203	-0.1611	Same
RA	F10	GAG disease / TGF-β	-0.2207	-0.2382	Same
RA	F11	BMP signalling	-0.1996	0.1436	Opp.
RA	F15	NTRK2 / TrkB	-0.3853	-0.1856	Same
RA	F16	Keratan sulfate	0.0841	-0.0315	Opp.

No significance encoding.
Regional sign: Same = shared LAA/LAPV sign; Opp. = opposite.
Chronicity classes use unrounded LAA Spearman ρ.

−1.0 −0.5 0.0 0.5 1.0
negative positive
Spearman ρ with AF duration (−1 to +1)

C Chronicity classes

LAA |ρ| bins across 16 frozen NCP families

2	6	8

Strong-trend 2/16
$|\rho_{LAA}| \geq 0.4$
F2 · F9

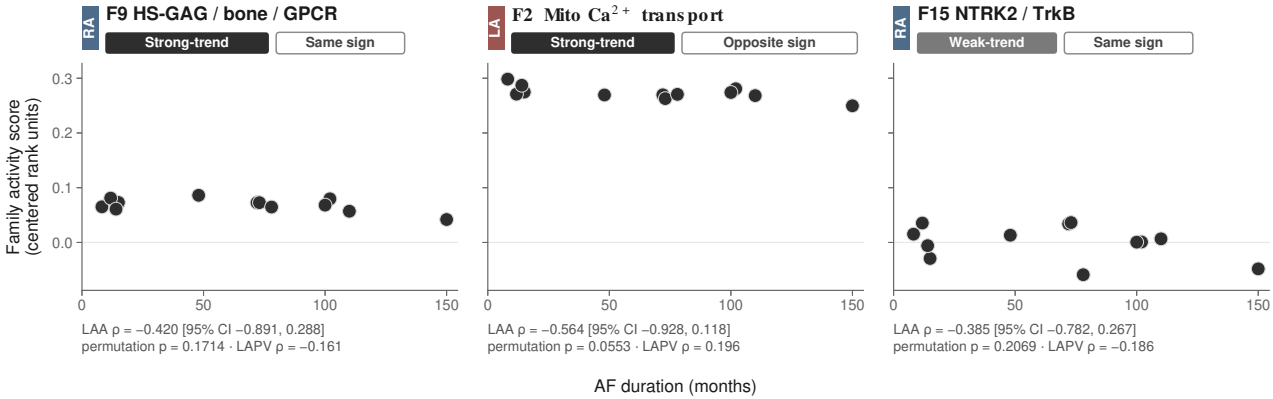
Weak-trend 6/16
$0.2 \leq |\rho_{LAA}| < 0.4$
F3 · F7 · F8
F13 · F10 · F15

Stable 8/16
$|\rho_{LAA}| < 0.2$
F1 · F4 · F5 · F6
F12 · F14 · F11 · F16

Classes use unrounded $|\rho_{LAA}|$.
Direction shown in B.
Descriptive bins, not significance calls.

D Representative LAA duration trends and regional sensitivity

Strong and Weak LAA trends with concordant or discordant LAPV signs



12 AF-LAA samples per facet; classes use full-precision |ρ_LAA|;
LAPV is regional sensitivity only.